# GENERATIVE MODELLING

## GENERATIVE STORY

**Step 1:** Decide the label by tossing a coin.
$P(y_i = 1) = p$

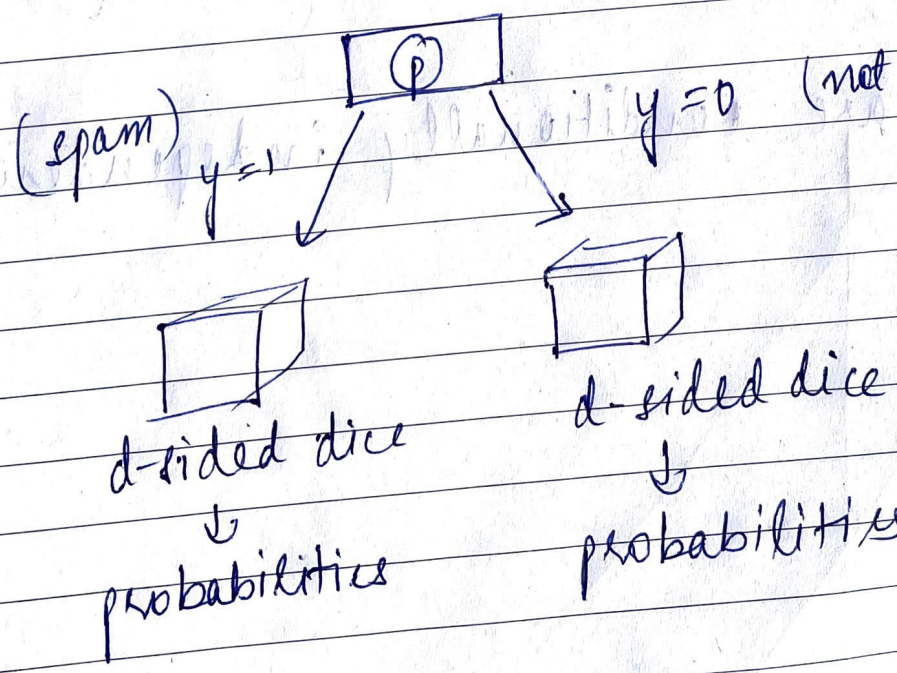
**Step 2:** Decide features using the label in step 1 by $P(x_i | y_i)$

$$\text{Number of parameters} = \underbrace{1}_{\text{to decide label}} + \underbrace{(2^d - 1)}_{P(x|y=1)} + \underbrace{(2^d - 1)}_{P(x|y=0)}$$

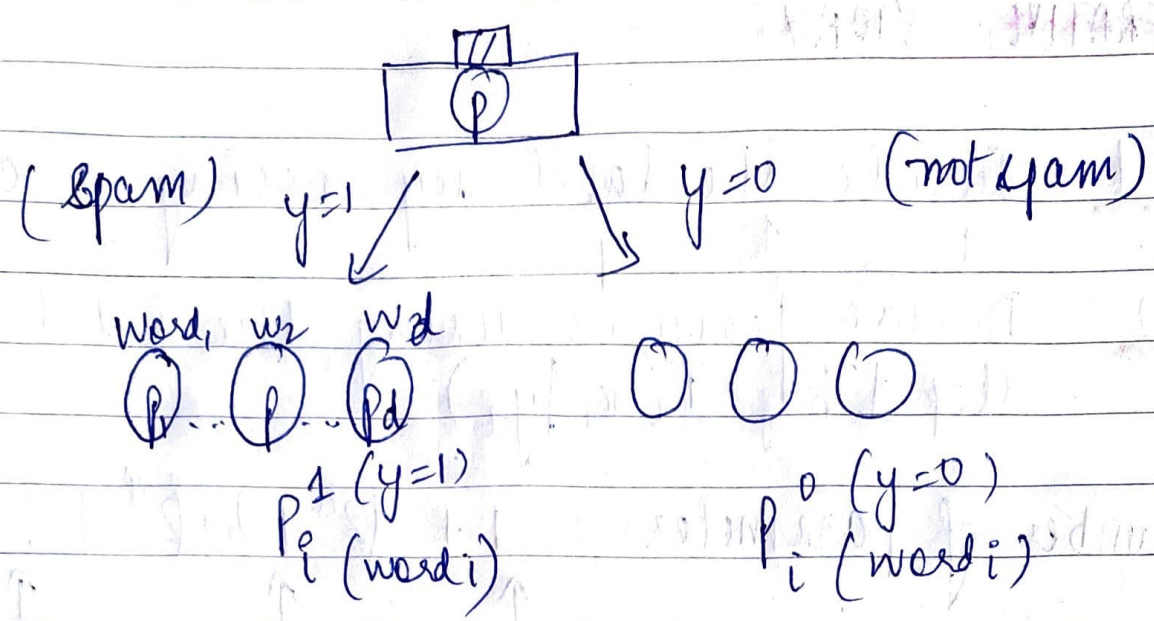
$$= 1 + 2(2^d + 1)$$

$$= 2^{d+1} - 1$$

Issues → - too many parameters
- not a reasonable story



# ALTERNATE GENERATIVE STORY




Number of parameters = $2d + 1$

**Step 1:** $P(y=1) = p$

**Step 2:** $P(x = [f_1\ f_2 \dots f_d] \mid y) =$

$$\prod_{i=1}^{d} \left(p_i^y\right)^{f_i} \left(1 - p_i^y\right)^{(1-f_i)}$$

Features are conditionally independent given label.

Date:
No:

# Parameter Estimation

To estimate : $p$, $\{p_1^1, \ldots p_d^1\}$, $\{p_1^0, \ldots p_d^0\}$

Maximum Likelihood Estimates:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^{n} y_i$$

$\forall j \in \{1, \ldots, d\}$ $\forall y \in \{0, 1\}$

$$\hat{p}_j^y = \frac{\sum_{i=1}^{n} \mathbb{1}(f_j = 1, y_i = y)}{\sum_{i=1}^{n} \mathbb{1}(y_i = y)}$$

# Prediction

Given $x^{test} \in \{0,1\}^d$, what is $\hat{y}^{test}$?

$$P(y^{test} = 1 \mid x^{test}) > P(y^{test} = 0 \mid x^{test})$$
$$\Rightarrow \hat{y}^{test} = 1$$

# Bayes Rule

$$P(y^{test} = 1 \mid x^{test}) = \frac{P(x^{test} \mid y^{test} = 1) \cdot P(y^{test} = 1)}{P(x^{test})}$$

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## Pitfalls in Naive Bayes Algorithm

If a point is not in the training set but appears in the test set, probabilities become zero irrespective of the fact the set may contain highly probable data points!

↳ add two 'pseudo' features with all words present - one has label 0 and the other has label 1

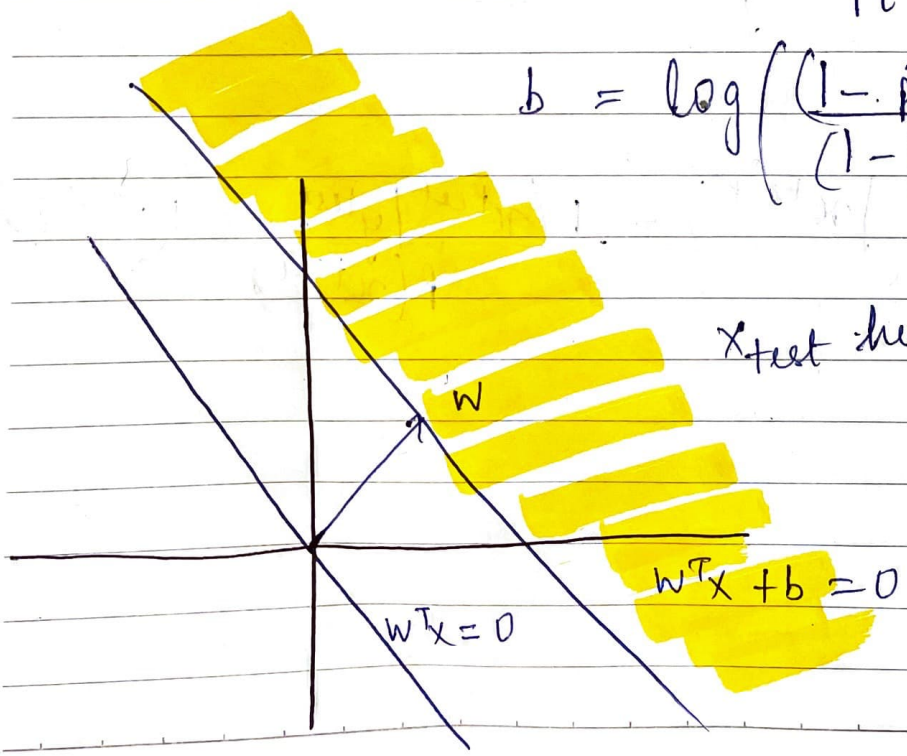
"Laplace smoothing"

## Decision function of Naive Bayes

Predict $y_{test} = 1$ if $\underset{\in \mathbb{R}^d}{\underset{\downarrow}{w^T}} x_{test} + b \geq 0$,

where $w_i = \log\left(\frac{\hat{p}_i^1 (1-\hat{p}_j^0)}{\hat{p}_i^0 (1-\hat{p}_i^1)}\right)$ and

$$b = \log\left(\frac{(1-\hat{p}_i^1)}{(1-\hat{p}_i^0)}\right) + \log\left(\frac{\hat{p}}{1-\hat{p}}\right)$$




$x_{test}$ here is 1

'Linear'

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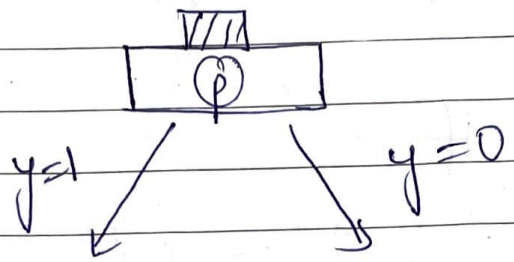
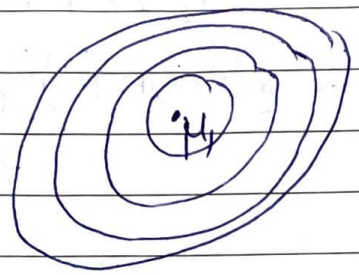
## Decision Boundary - Naive Bayes

$$\{x : P(y=1|x) = P(y=0|x)\}$$

## GAUSSIAN NAIVE BAYES




Illy

$$x|y=0 \sim N(\mu_0, \Sigma)$$

$$x|y=1 \sim N(\mu_1, \Sigma)$$

<u>Note</u>: In this model, covariances are assumed to be the same.

## Maximum Likelihood Estimates

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} y_i$$ → fraction of points labelled 1

$$\hat{\mu}_1 = \frac{\sum_{i=1}^{n} \mathbb{1}(y_i = 1) \cdot x_i}{\sum_{i=1}^{n} \mathbb{1}(y_i = 1)}$$ → Sample mean of data points labelled 1

$$\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n} (x_i - \hat{\mu}_{y_i})(x_i - \hat{\mu}_{y_i})^T$$

## Prediction

$$P(y_{test} | x_{test}) \propto \underbrace{P(x_{test} | y_{test})}_{f(x_{test};\, \hat{\mu}_{y_{test}},\, \hat{\Sigma})} \cdot \underbrace{P(y_{test})}_{\hat{p}}$$

Predict $y_{test} = 1$ if

$$f(x_{test}; \hat{\mu}_1, \hat{\Sigma}) \cdot \hat{p} \geq f(x_{test}; \hat{\mu}_0, \hat{\Sigma}) \cdot (1-\hat{p})$$

$$e^{-(x_{test}-\hat{\mu}_1)^T \hat{\Sigma}^{-1} (x_{test}-\hat{\mu})} \hat{p} \geq e^{(x_{test}-\hat{\mu}_0)^T (x_{test}-\hat{\mu}_0)} (1-\hat{p})$$

On simplification,

Predict 1 if,

$$\underbrace{\left((\hat{\mu}_1 - \hat{\mu}_0)^T \hat{\Sigma}^{-1}\right) x_{test}}_{w^T x_{test}} + \hat{\mu}_0^T \hat{\Sigma}^{-1} \hat{\mu}_0 - \hat{\mu}_1^T \hat{\Sigma}^{-1} \hat{\mu}_1 + \log\left(\frac{1-\hat{p}}{\hat{p}}\right) \geq 0$$

$$w^T x_{test} + b \geq 0$$

Decision function is linear.

for same $\Sigma$